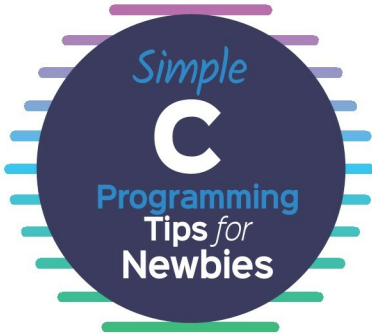




Many programmers begin their romance with code by learning C. The author suggests a few tips for newbies so they can avoid bad coding practices.

The C programming language is usually the first one that students are taught in most colleges. It is one of the most widely used languages, particularly in systems programming. Here are a few simple tips to avoid some of the most common pitfalls in the language.

The *scanf* function and character input

One common error is due to the characters remaining in the keyboard buffer after an input is executed by a *scanf* statement.

When there are two or more consecutive *scanf* statements in the program and the second or later *scanf* statement is for character input, it looks as if the *scanf* statement is skipped. This is because the newline character, *\n*, which was input due to the return key being pressed by the user, remains in the keyboard buffer. If the next *scanf* is for character input, the newline character is taken as the input for the character variable.

```
scanf("%d", &num);
scanf ("%c", &chr);
```

This does not happen for the integer input, as the *%d* symbol for integers forces all leading white spaces and special characters in the input stream to be skipped.

This situation can be avoided by using a space before *%c*. The space character will ensure that the *scanf* statement skips the leading white spaces and special characters, and the above

problem can be avoided.

```
scanf("%d", &num);
scanf (" %c", &chr); /*space character added before %c */
```

The *fflush* function

The *fflush* function can also be used for flushing the keyboard buffer.

```
fflush (stdin);
```

But the *fflush* function is avoided by many as it is undefined in the C standard for the input stream.

Magic numbers

When a number suddenly appears in some part of a program without any background or explanation, it is said to be a magic number. The introduction of magic numbers is a result of poor programming practices and also due to writing a program without thinking it through.

For example:

```
for (i=0; i<10; i++)
    scanf ("%d", &a[i]);
```

The above code is used to input an array of integers of size 10. The number 10 has to be used repeatedly to mention